UNIT 3: CELLS
3.1 – CELL BIOLOGY

Biology 9   Mr. Queenan

Chapter 8.1 (p.238-247) & Ch 11.1 (p.339)

Unit Essential Question

How does structure relate to function on a cellular level?

Section 1: The History of Cell Biology

History of Cells

- All living things are made of one or more cells
  - Smallest unit that can carry on all processes of life
- As early as the **17th century**, scientists used **microscopes** to observe objects too small to be seen by eye and develop the idea of **cell-based life**
  - **Robert Hooke**: observed thin slices of cork with early microscope; "little boxes" observed reminded him of cubicles or "cells" where monks live
    - Observed similar structures in sections of tree stems, roots & ferns
  - **Anton van Leeuwenhoek**: first to observe living cells, including microorganisms such as algae Spirogyra & Vorticella
    - Called microorganisms "animalcules"; now known as protists

Cell Theory

- Summary of work of Hooke, Leeuwenhoek, Schleiden, Schwann, and Virchow from 16-1900s
  - 1: All living organisms are composed of one or more cells
  - 2: Cells are the basic units of structure and function in an organism
  - 3: Cells come only from the reproduction of existing cells (self-reproducing)

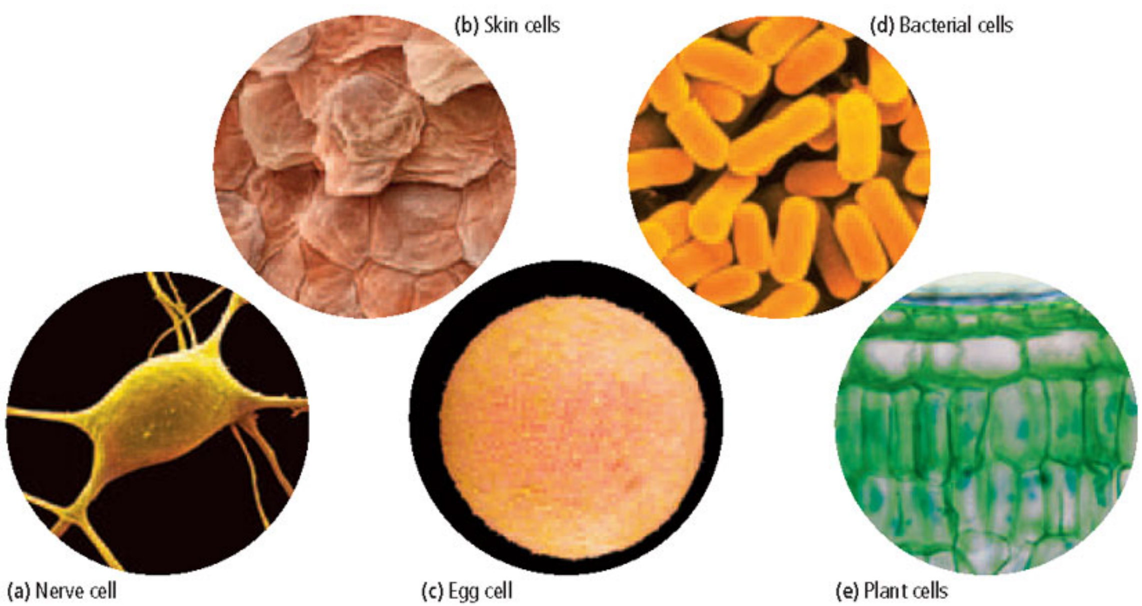
Contributions of Microscopy

- Study of cell biology limited until development of the **compound light microscope**
  - Continues to expand with development of **electron microscopes** and other **molecular biology tools**
- Microscopy lead to clarifications of our definition of life, that **all living things share basic characteristics**:
  - Consist of organized parts
  - Obtain energy from surroundings
  - Perform chemical reactions
  - Change with time
  - Respond to environment
  - Reproduce
  - Separate their constant internal environment from the changing external environment
  - Share a common history – common characteristics that show how cells are related to other living things

Section 2: Introduction to Cells

Diversity of Cells

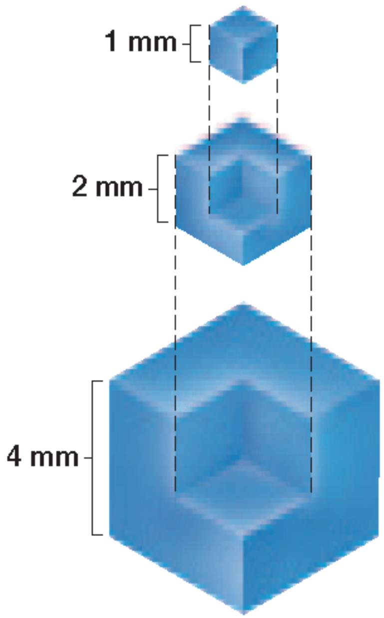
- Cells, both among and within, organisms are very diverse:
  - Vary in **shape, size, internal organization, function**
- Varying function of cells a product of their different structures
  - Can be **simple** or **complex** depending on function
  - Structure has **evolved** to allow most effective **functionality**



Size of Cells

- Most cells too small to be seen with the naked eye
  - **10-50 µm** in diameter
  - Except some nerve cells and human egg cell
- Cell size **limited** by relationship between **surface area** and **volume** (surface area-to-volume ratio)
- During cell growth, **volume** increases much **faster** than **surface area**
  - Necessary materials (nutrients & O_2) and waste products (CO_2) must pass through cell surface
  - Cell size limited to **microscopic level** because **surface area** needs to be **sufficient** to support needs of cell

Size of Cells

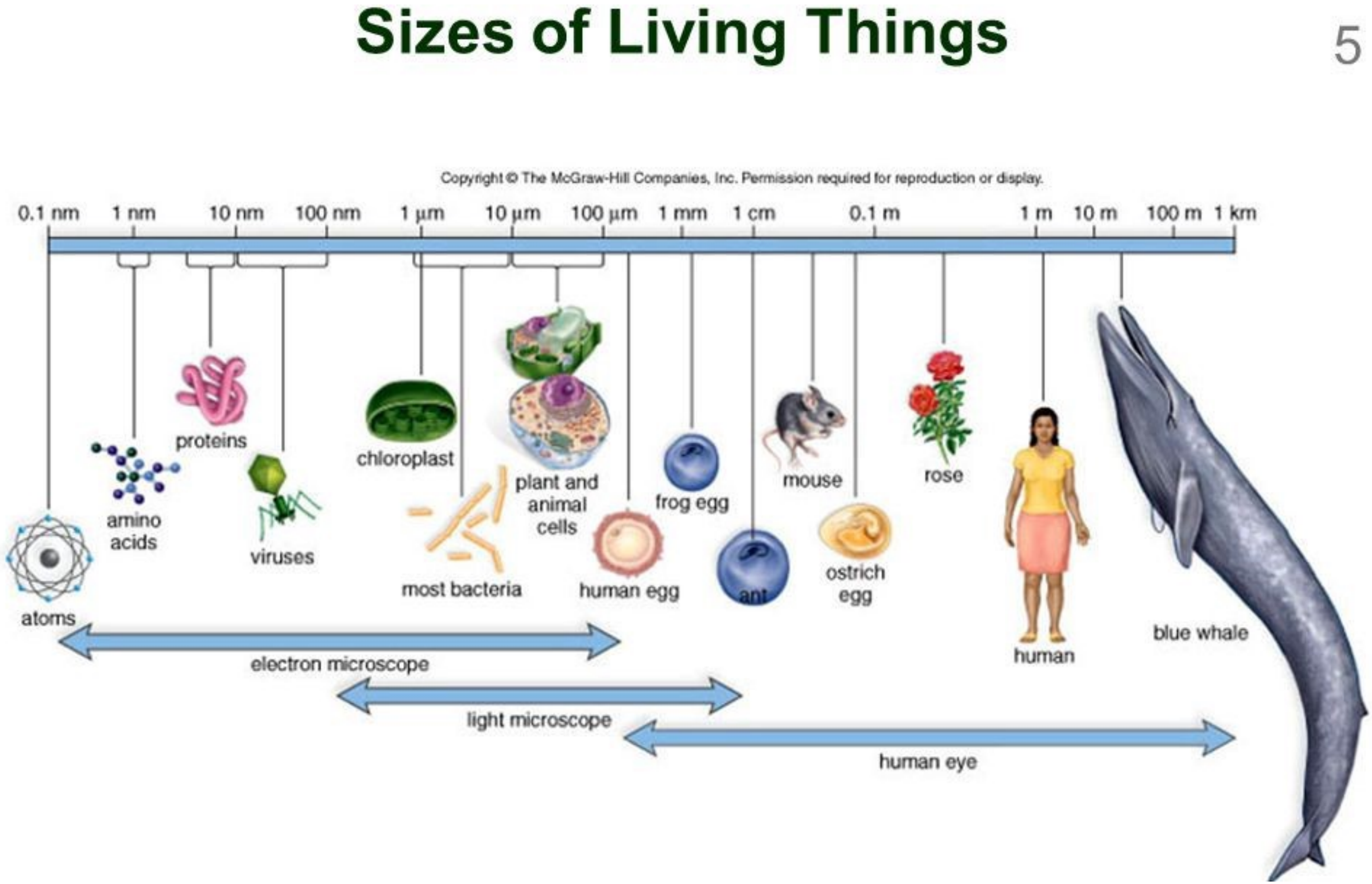


Side length	Surface area	Volume	Surface area/ volume ratio
1 mm	6 mm^2	1 mm^3	6:1
2 mm	24 mm^2	8 mm^3	3:1
4 mm	96 mm^2	64 mm^3	3:2

Smaller cell → higher surface area : volume ratio

Need more surface are to allow adequate nutrients to enter and waste to be expelled

Size and Scale



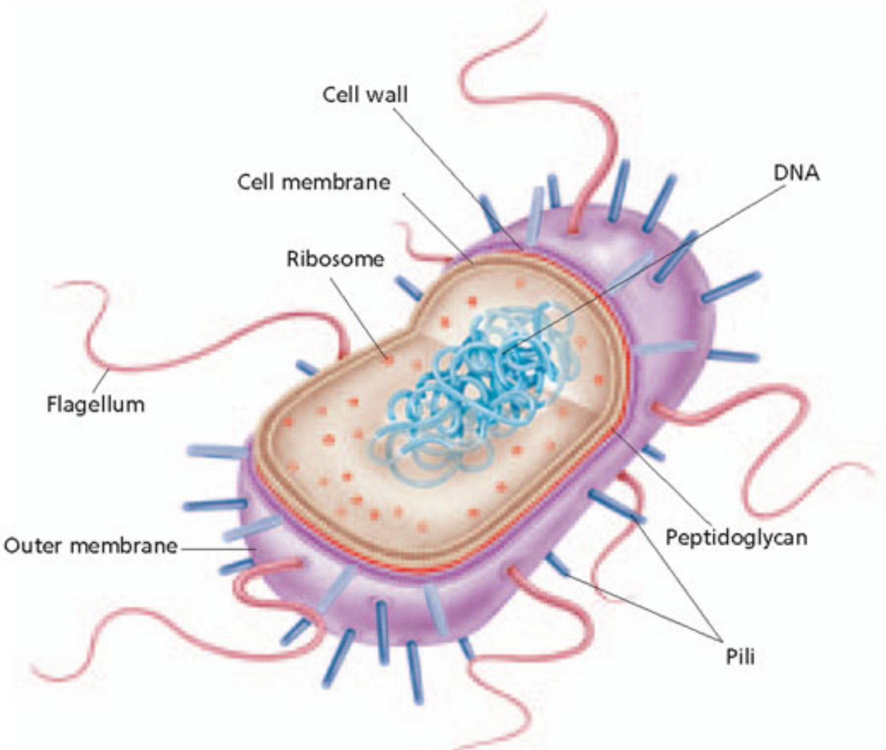
Basic Parts of the Cell

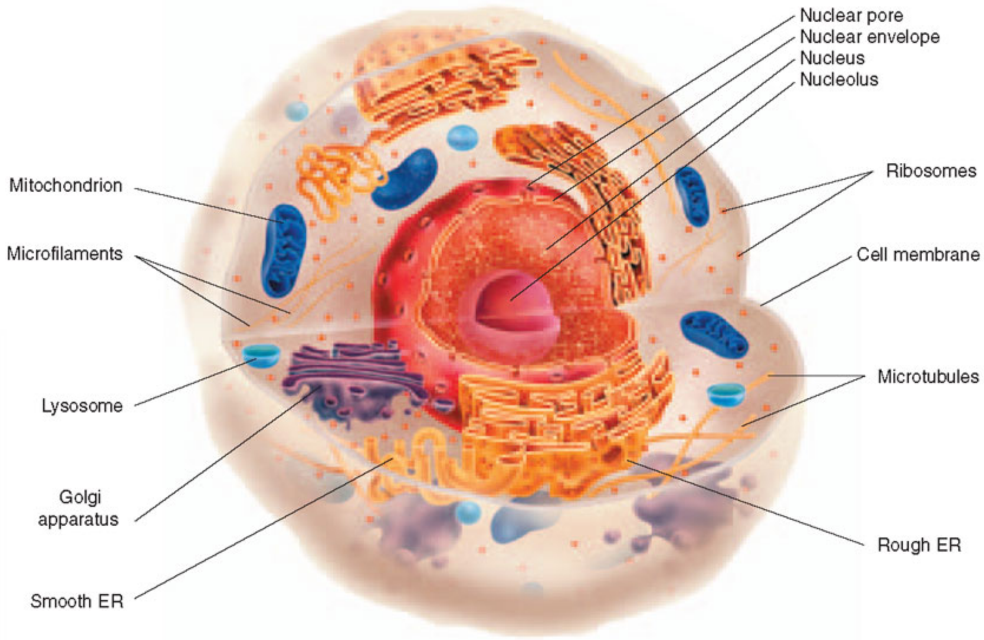
- **Plasma Membrane:** cell's outer boundary
  - Cover's the cell's surface and acts as a **barrier**
  - Materials **enter** and **exit** through plasma membrane
- **Cytoplasm:** region of the cell within the plasma membrane
  - Includes **fluid, cytoskeleton**, and all **organelles** except nucleus
  - **Cytosol:** part of cytoplasm that includes **molecules, small particles (i.e. ribosomes)**
    - Composed of ~**20% protein**
- **Control Center:** membrane-bound organelle that contains the cellular DNA (**nucleus**)
  - **Controls** most **functions** of the cell
  - **Largest cell structure**; easiest to see
  - Maintains its shape via **protein skeleton** known as **nuclear matrix**

Basic Types of Cells

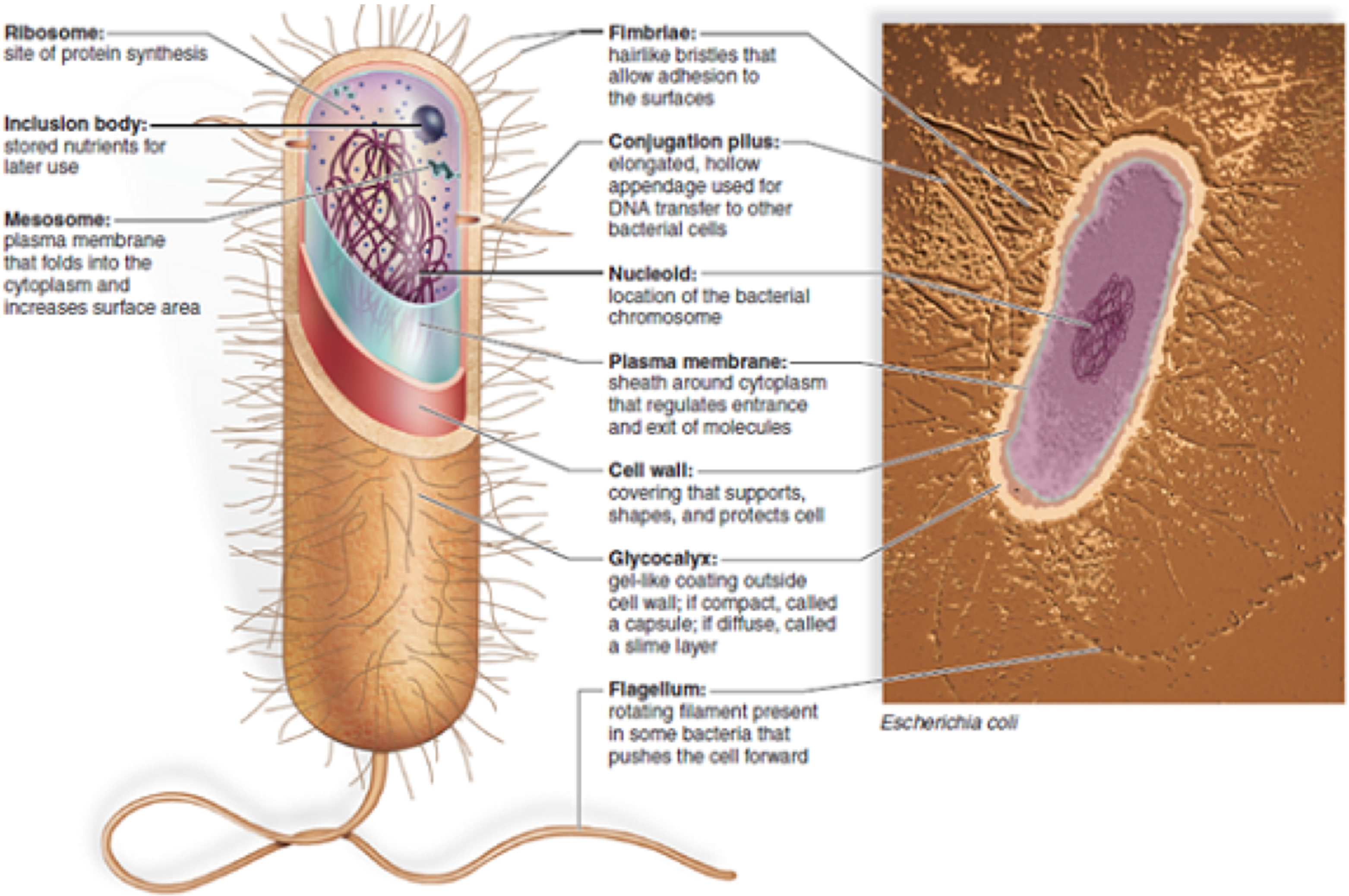
- 2 basic **types** of cells:
  - **Prokaryotes**: organisms that **lack** membrane-bound nucleus and membrane-bound organelles
    - DNA concentrated in an area called **nucleoid**
    - Divided into 2 Domains: **Bacteria** & **Archaea**
      - **Bacteria**: organisms that are similar to **first cellular life forms**
      - **Archaea**: organisms **more closely related to eukaryotic cells**
  - **Eukaryotes**: organisms composed of one or more cells that **have** a nucleus & membrane-bound organelles
    - **Organelle**: sub-cellular structure; well-defined, intracellular bodies that perform specific functions for the cell
      - Perform functions for the cell as organs perform functions for the body
    - Cells generally **larger** than those of prokaryotes

Basics of the Cell





Prokaryotic Differences



Cellular Organization

- Some cells can live alone; others depend on presence of other cells to survive
- **Colonial organism**: collection of genetically identical cells that live together in a connected group. Ex from lab activity – algae, volvox
  - Not truly multicellular because few cell activities are coordinated
- **Multicellularity**: an organism composed of multiple cells that are **specialized** in their function and **unable to survive independently**
  - **Tissue**: group of **similar cells** that perform a particular job in an organism
  - **Organ**: group of **similar tissues** that perform a particular job in an organism
  - **Organ system**: group of **organs** that accomplish related tasks